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Understanding Student Loan Debts Across U.S. Universities

With college tuition increasing every year, many incoming students question whether gaining a higher education is worth the potential financial burden. This project analyzes the relationship between the cost of tuition, how universities are ranked, and student loan debt at universities across the United States. One big question is whether or not attending a more prestigious college will leave a student with more debt than it may be worth. We will be comparing two datasets, both with credible and real-world information.

Both datasets were found on the website Kaggle. The first dataset is titled “The Schools that Create the Most Student Debt,” from the Common Origination and Disbursement System through the Department of Education. The second dataset, titled “2022 US College Rankings and Costs,” contains information from the U.S. News & World Report. The student debt dataset includes details about median student debt from over 40,000 schools, with columns containing school, city, state, zip code, school type, loan type, recipients, number of loans originated, amount of money originated in loans, number of disbursements, and amount of money disbursed. The college ranking dataset only contains 160 schools, but this will make it easier for us to analyze specifically the more prestigious colleges. The columns in this dataset contain a school’s name, adjusted rank, tuition, and enrollment numbers. Together, these datasets cover a large range of private and public colleges and universities across the country.

While conducting this project, we understand that there may be confounding variables that affect student loan debt that are not completely explained by the cost of tuition and school ranking. There may be factors such as family income, financial aid, or cost of living that could also influence student loan debt. Our current datasets do not include these variables, but we plan to recognize these limitations throughout our process.

Materials and Methods: We started by cleaning and merging the two datasets. The student loan dataset included information for over 40,000 schools, while the college ranking dataset included 160 universities. After merging, we had to remove entries with missing values. Schools with missing rank, tuition, enrollment numbers, or loan information were removed from the dataset. We then aggregated the data by school and created new variables to look at total and average loan information. Our analysis focused on exploring relationships between school rank, tuition, average student debt amounts per student, and the different types of loans. We performed summary statistics and conducted correlation analyses between rank, tuition, and average student loan debt per student. By fitting linear regression models with rank or tuition separately and completing a Monte Carlo simulation, we were able to predict average student loan debt per student and in 10 years. Also, we created variables to measure loan efficiency and waste by subtracting total loan amounts from total disbursements. We ran hypothesis tests on different types of loans and were able to find standard errors.

Results: Summary statistics show that the values in the dataset range from approximately $17,000 to $63,000, with an average of about $46,500. School ranks range from 1 to 100, and average student loan debts per borrower range from about $5,000 to $33,000, with a mean of about $13,200. The correlation analysis revealed a statistically significant negative relationship between school rank and average student loan debt per student, and a statistically significant positive correlation between tuition and average student loan debt per student. This shows that higher tuition and better-ranked schools correspond to higher student loans per borrower.

A regression analysis was used to support these findings. We did three different models: model 1, using only rank as a predictor, model 2, using only tuition as a predictor, and model 3, using both rank and tuition as predictors. Model 1 showed us that 30.5% of the variation in student loan debt per student is explained by rank. Model 2 was also statistically significant and showed that tuition explains about 29.2% of the variation. Lastly, Model 3 showed that both rank and tuition together explain 37.2% of the variability. Also, with the regression analysis, we saw that the average loan amount per student decreases by about $45 for each one-place increase in rank. For every increase in tuition of $1, the average loan per student increases by $0.15. So, rank is the stronger factor, but tuition adds a meaningful contribution to explaining loan debts per student.

A Monte Carlo simulation was used to project the average student loan debt 10 years from when the data was collected, assuming tuition increases at 4% annually and the relationship between tuition and

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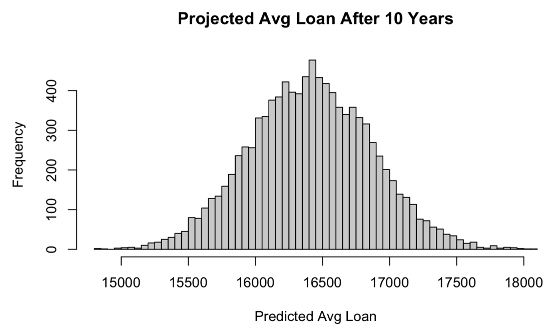
student loan debt stays the same. Using our linear regression model 2, we simulated 10,000 potential loan outcomes. The simulation predicted that after 10 years, the average student loan debt per borrower could rise to about $16,500, which is higher than the current observed average of $13,168. Based on the sample of schools in our dataset, the 95% confidence interval for the true mean average student loan debt is estimated to be between $12,022 and $14,315. This means that we are 95% confident that the true average student loan debt amount per student falls within this range.



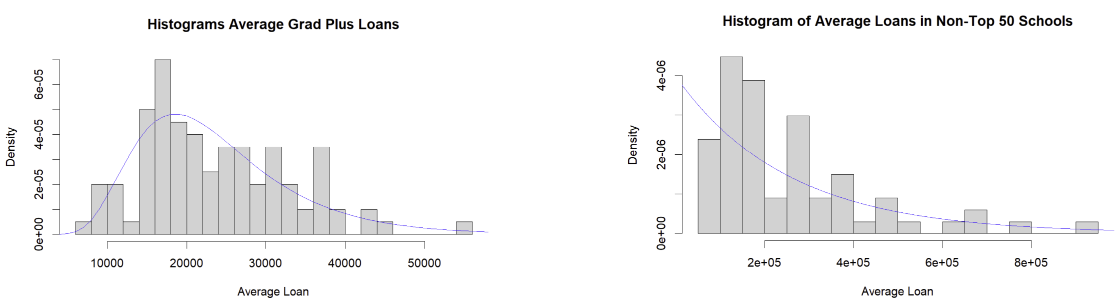
<u>Figure 1:</u> Simulated Loan amounts from the Monte Carlo simulation

A predictive calculation was created using the regression model to estimate the amount of average student loan debt and the 95% confidence interval when given a school's rank and tuition cost. We found that a school with a rank of 30 and tuition of $60,000 would have an average student loan debt of about $17,408, with a 95% chance that it falls between about $8,000 and $26,815.

To find out how efficient the student loans are, we subtracted total loan amounts from total disbursements and saw how much money from loans was not being used by each school. We found the top 10 schools that have the most loan waste, the top one being Arizona State University, with a loan waste of almost $7.4 billion. This number is almost 2-3 billion dollars higher than the other 9 top schools, and this could be because of the incredibly high enrollment number at ASU, considering they have over 70,000 students. Many students are approved for loans, but do not end up using the entire amount.

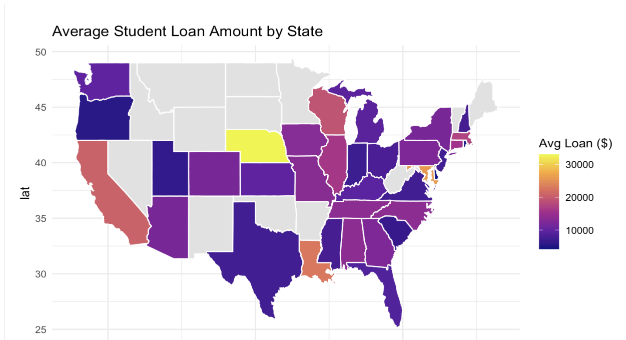
We also ran 2 hypothesis tests, and we found them both to be true. First, we found that the mean "Grad Plus" loan is greater than the mean "Parent Plus" loan. Then, we also found that there is no difference between the average loan in the top 50 schools vs. the non-top 50 schools. We wanted to test that the empirical type 1 error rate of 50 simulated loans from the most similar distribution to our "Grad Plus" loan distribution would still be greater than our average "Parent Plus" loan, which was about \$26,780. After running our tests, we found that the empirical type 1 error rate for our test was 0.014 with a standard error of 0.003, meaning our type 1 error rate was controlled at the 95% confidence level, so we assume 50 new "Grad Plus" loans will, on average, be larger than "Parent Plus" loans.

We ran the same sort of test on our non-top 50 school loans vs our top 50 school loans to test the type 1 error rate on them being equal, based on 50 simulated loans from non-top 50 schools. Our Mean value for the top 50 school loans was \$25,400. After running our simulations, we found a type 1 error rate of 0.027 with a standard error of 0.005, meaning our type 1 error rate was controlled at the 95% confidence level, so we assume the average of 50 non-top 50 school loans will be the same as the top 50 school loans.



<u>Figure 2:</u> Showing the distributions and curves of the tested loans

We were also able to create a map that shows the average student loan by state on a map of the United States. We did this by taking the data, merging by state, and creating a brand new column that finds the average loan distribution per student. After doing this, we were able to use online sources to help install a "usmap" in R and then merge the downloaded map into our data. After making minor adjustments to the data, we were able to plot the merged data and usmap (seen in Figure 3) together and make a legend with colors to easily identify the average student loan distribution per student in every state where we have data.



<u>Figure 3:</u> US Map of Average Loan Distribution per student in each state

Our analysis of student loan debt across many Universities all over the United States reveals a clear relationship between tuition costs, school rankings, and the average loans per student. Through regression analysis, Monte Carlo simulations, and hypothesis testing, we have found evidence that tuition and rank have an impact on student loan debt. Despite some limitations, such as a lack of variables (family income, cost of living, financial aid), our study provides a good foundation for understanding the financial expectations of choosing colleges.

For the project's future, getting a larger rank dataset could give better and more accurate results. Some questions would be how focusing on median student loan debts (which would reduce the effect of extreme outliers) would alter the results. Our findings can help prospective students make informed financial decisions and understand the costs of attending college.